Testing and Verifying Local Deployment

Milestone 5 - Activities 5.1 and 5.2

1. Overview

Milestone 5 is about running ComicCraft on your own computer to confirm the whole workflow works: AI text generation, image creation, layout building and PDF export. It has two activities: **5.1 Preparing the Application for Local Deployment** and **5.2 Testing and Verifying Local Deployment**. This document covers both, since 5.2 depends on 5.1. Local testing also prepares the app for a possible cloud deployment in the future.

2. Activity 5.1 - Preparing for local deployment

Step 1 - Set up a virtual environment

A virtual environment isolates the project's dependencies from other Python projects. Create it, activate it, then install everything from requirements.txt (FastAPI, Uvicorn, Google Generative AI, Hugging Face Diffusers, FPDF and others).

```
python -m venv env

# For Windows
env\Scripts\activate

# For macOS/Linux
source env/bin/activate

pip install -r requirements.txt
```

Step 2 - Configure environment variables

Sensitive values such as the Gemini API key and the Hugging Face token should never be written directly in the code. Set them in the terminal, or place them in a `.env` file that the app loads using **python-dotenv**:

```
GEMINI_API_KEY=your_gemini_api_key_here
HF_API_KEY=your-huggingface-api-key-here
```

Important: Do not upload the .env file to Git or share it. If you use Git, add .env to .gitignore.

Step 3 - Templates via Jinja2

All user interactions are handled through Jinja2-rendered templates served by FastAPI. Make sure the templates folder and the static folders (static/panels for images, static/exports for PDFs) exist before starting. The main pages of the working app are index.html (input form), comic_preview.html (result) and export_success.html (confirmation).

3. Activity 5.2 - Testing and verifying

Step 1 - Start the FastAPI server

Go to the project root, where `app/main.py` is located, and start Uvicorn with hot reload so the server restarts whenever you edit code:

```
uvicorn app.main:app --reload
```

Step 2 - Open the application

In any browser, open `http://127.0.0.1:8000`. This loads the ComicCraft homepage.

Step 3 - Test the API docs

FastAPI provides interactive documentation at `http://127.0.0.1:8000/docs`. From here you can try the endpoints directly: `/generate-comic/json`, `/generate`, `/export-success` and `/test-image`. Using /test-image first is a quick way to confirm Stable Diffusion works without running the whole pipeline.

4. Walkthrough of the website pages

Home page

A clean starting page with a scenic background. A form in the centre collects the inputs for comic generation:

- **Story Prompt** - your comic idea.
- **Main Character Name**.
- **Setting** - school, forest, space, city.
- **Story Tone** - light-hearted, dramatic, poetic, funny.
- **Art Style** - anime, pixel art, comic book, realistic.

On submit, the AI creates the outline, dialogues, illustrations and an exportable PDF.

Comic Preview (result) page

Shows the generated panels one after another. Each panel contains:

- **Panel title** with number, e.g. "Panel 2: Into the Deep Woods".
- **Comic image** generated from the scene's image prompt.
- **Scene description** in italics, setting the mood.
- **Caption and narration** - ambient description plus the character's actions, emotions or dialogue.
- **Image prompt reference** - the text used to generate the picture.

At this stage the page is only for viewing and reviewing the story before exporting.

Download button

The "Download Your Comic as PDF" button compiles images, titles, descriptions, captions and narration into a formatted PDF and downloads it to the device, so it can be saved, shared or viewed offline.

Export Success page

After the download is triggered, the user is redirected here. It confirms the comic was created and downloaded, and offers a "Go Create Another Comic" button that returns to the homepage.

5. Testing checklist

#	What to verify	Expected result
1	Server starts with uvicorn app.main:app --reload	No errors in the terminal; server listening on port 8000
2	Open http://127.0.0.1:8000	Homepage with the input form loads
3	Open /docs	Swagger UI lists /generate, /generate-comic/json, /export-success, /test-image
4	Call /test-image with a sample prompt	An image is generated and saved in static/panels
5	Submit the form with all five fields	Comic Preview page shows 5 panels with images and text
6	Change tone and art style and submit again	Story mood and image style change accordingly
7	Click Download Your Comic as PDF	PDF downloads; a file appears in static/exports
8	After download	Redirected to the Export Success page
9	Click Go Create Another Comic	Returns to the homepage

6. Common problems and quick fixes

These are general troubleshooting tips (not from the project document):

Problem	Things to check
Module not found error	Is the virtual environment activated? Re-run pip install -r requirements.txt.
Gemini API error	Is GEMINI_API_KEY set correctly and loaded from .env? Restart the server after changing it.
Image generation very slow or out of memory	Stable Diffusion is heavy; on CPU it is slow. Close other apps or use a GPU if available.
Form submits but nothing happens or 422 error	Field names in index.html must match the Form(...) parameter names; python-multipart must be installed.
Port already in use	Stop the other process or run Uvicorn with a different port using --port.
Images or PDF not visible	Check that static/panels and static/exports exist and static files are mounted.

7. Conclusion

When all checks in the list pass, the local deployment is verified: the full path from user input to AI generation, preview, PDF export and success confirmation works on your machine. The app is then ready for future improvements such as user profiles, a comic library or a cloud deployment.